QUANTUM ALPHA

Reinforcement Learning Strategy for Dynamic Portfolio Optimization During Market Volatility

Research Paper

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Project Focus:

NIFTY 50 Index | Regime-Aware Reinforcement Learning | Deep Q-Network (DQN)

Research Framework:

Reinforcement Learning | DQN Architecture | Custom Trading Environment |

Walk-Forward Validation | Regime Analysis | Stress Testing

April 2026

DECLARATION

I, Ayush Rawat, a student of BBA (Finance) at Tecnia Institute of Advanced Studies, Guru Gobind Singh Indraprastha University, Delhi, hereby declare that the research paper titled

"Quantum Alpha: Reinforcement Learning Strategy for Dynamic Portfolio Optimization During Market Volatility" is my original work carried out under the guidance of the faculty. This work has not been submitted in part or full for the award of any degree, diploma, or any other academic qualification to any other university or institution.

I further declare that all sources of information used in this research have been duly acknowledged.

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Date: **3 April 2026**

Abstract

This study introduces **Quantum Alpha**, a Deep Q-Network (DQN) based reinforcement learning framework designed for dynamic long/short/flat portfolio positioning in the Indian equity market (NIFTY 50). The agent learns optimal exposure decisions using a rich 7-dimensional state representation that incorporates momentum, volatility, trend strength, drawdown, VIX, and market breadth signals.

Trained on daily data from 2019 to 2022 and evaluated out-of-sample on 2023–2025, the RL agent demonstrates superior risk management compared to a traditional momentum rule-based benchmark. While achieving a negative CAGR of -3.43%, the model significantly reduces maximum drawdown to -15.07% (versus -31.80% for the rule-based strategy) and volatility to 6.12%. The agent also exhibits clear regime-adaptive behaviour, increasing defensive positioning during high-volatility periods.

These findings highlight both the potential and current limitations of deep reinforcement learning for emerging market trading, particularly the challenge of balancing risk control with return generation in strong bull markets. The project contributes to the growing literature on regime-aware RL applications in volatile financial environments and provides a complete end-to-end framework for future research in emerging markets.

Keywords: Reinforcement Learning, Deep Q-Network, Portfolio Optimization, Market Regimes, Emerging Markets, Risk Management, NIFTY 50

EXECUTIVE SUMMARY

This research presents **Quantum Alpha**, a Deep Q-Network (DQN) reinforcement learning framework developed for dynamic long/short/flat portfolio positioning in the highly volatile Indian equity market (NIFTY 50). The agent operates with a 7-dimensional state representation incorporating momentum, volatility, trend persistence, drawdown, India VIX signals, and market breadth, enabling implicit regime-aware decision-making.

Trained on daily data from 2019 to 2022 and evaluated out-of-sample over 2023–2025, the system demonstrates clear superiority in risk management relative to a traditional momentum-based benchmark. The RL agent achieved a maximum drawdown of -15.07% (versus -31.80% for the rule-based strategy) and reduced annualised volatility to 6.12% (versus 10.25%). These improvements validate the effectiveness of reinforcement learning in delivering robust downside protection during volatile emerging market conditions.

Notably, the agent exhibited strong regime-adaptive behaviour, increasing defensive (flat) positioning during high-volatility periods while maintaining controlled exposure across bull and bear regimes. However, the model remained conservative, with approximately 67% flat exposure, resulting in negative overall returns and a Sharpe ratio of -1.60. This outcome reflects a deliberate trade-off in the reward function design, prioritising capital preservation over aggressive return generation.

The study contributes a complete, reproducible end-to-end pipeline — including custom Gym-style environment design, advanced reward shaping, walk-forward validation, multi-regime stress testing, and robustness analysis — that addresses key challenges in applying deep reinforcement learning to real-world emerging market trading. While the current implementation highlights the difficulty of balancing risk control with upside capture, it establishes a solid foundation for next-generation adaptive portfolio systems.

Keywords: Reinforcement Learning, Deep Q-Network, Regime-Aware Trading, Emerging Markets, Risk Management, Portfolio Optimization

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Without the combined support of my mentors, family, and modern AI tools, this ambitious research would not have reached its current standard.

Ayush Rawat

BBA (Finance)

April 3

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1. Introduction

The Indian equity market, represented by the NIFTY 50 index, is one of the fastest-growing and most volatile emerging markets globally. Characterised by frequent regime shifts, sharp drawdowns during global shocks, and prolonged periods of uncertainty, the NIFTY has repeatedly demonstrated the limitations of traditional investment approaches. Buy-and-hold strategies suffer significant losses during crashes, while simple momentum or trend-following rules often generate false signals in sideways or high-volatility regimes.

In such an environment, static or rule-based portfolio strategies frequently fail to protect capital or capture opportunities effectively. This has created a strong need for adaptive, dynamic portfolio optimization systems that can adjust exposure in real time — moving between long, short, or flat positions based on prevailing market conditions.

Reinforcement Learning (RL), particularly Deep Q-Networks (DQN), has emerged as a promising approach to address this challenge. Unlike traditional models that rely on fixed rules or historical averages, RL agents learn optimal decision-making policies through continuous interaction with the market environment. By treating trading as a sequential decision problem, RL can potentially capture complex, non-linear patterns and adapt to changing market regimes.

However, applying RL to real-world trading remains difficult. Key challenges include designing effective reward functions, managing transaction costs, avoiding degenerate policies (such as 100% flat positioning), and ensuring robustness in out-of-sample and stress-test conditions. Most existing studies focus on developed markets (e.g., S&P 500), leaving emerging markets like India relatively under-explored.

This research addresses these gaps by developing **Quantum Alpha** — a Deep Q-Network based reinforcement learning agent specifically designed for the NIFTY index. The agent learns to dynamically manage long, short, and flat exposure using a rich state representation that incorporates momentum, volatility, trend strength, drawdown, VIX behaviour, and market breadth signals.

Research Objective To design, train, and rigorously evaluate a DQN agent that adapts to volatility shocks and regime changes in the NIFTY market, with the goal of achieving superior risk-adjusted performance (measured by Sharpe, Sortino, Calmar, and Maximum Drawdown) compared to a traditional momentum rule-based benchmark.

The study contributes to the growing literature on reinforcement learning in emerging market finance by providing a complete end-to-end framework, including environment design, reward shaping, walk-forward validation, multi-regime stress testing, and detailed robustness analysis.

The remainder of this paper is structured as follows: Section 2 reviews relevant literature, Section 3 describes the methodology, Section 4 presents the empirical results, Section 5 discusses findings and limitations, and Section 6 concludes with future research directions.

2. Literature Review

Reinforcement Learning (RL) has gained significant traction in quantitative finance due to its ability to handle sequential decision-making under uncertainty. Early applications focused on simple Q-learning for portfolio allocation (Moody & Saffell, 2001), while recent advances use Deep Q-Networks (DQN) to handle high-dimensional state spaces (Mnih et al., 2015).

In the context of equity trading, several studies have applied RL to dynamic asset allocation. Li et al. (2019) demonstrated that DQN agents can outperform traditional mean-variance optimization in volatile markets by learning optimal position sizing. Similarly, Buehler et al. (2019) showed that deep reinforcement learning agents can effectively manage transaction costs and market impact in high-frequency settings.

A key challenge in applying RL to trading is reward function design. Jiang et al. (2017) highlighted that naive return maximization often leads to unstable policies, while incorporating risk penalties (e.g., drawdown or volatility) improves robustness. This is consistent with our findings, where aggressive return scaling was required to overcome degenerate "do-nothing" policies.

Regime-aware RL is another emerging area. Zhang et al. (2021) proposed regime-conditioned agents that switch between different policies based on volatility clustering, showing superior performance during market crashes. Our work builds on these ideas by using normalized technical and macro signals as part of the state vector to enable implicit regime adaptation.

Key Gaps in Existing Literature Most studies focus on developed markets (S&P 500, Euro Stoxx); emerging markets like NIFTY remain under-explored. There is also limited research on explicit long/short/flat action spaces with realistic transaction costs in emerging markets. Finally, few papers provide comprehensive walk-forward validation and multi-regime stress testing. This project addresses these gaps by developing a regime-adaptive DQN agent specifically for the NIFTY index.

3. Methodology

This section details the research design, data preparation, reinforcement learning environment, model architecture, reward function, training procedure, and evaluation framework implemented in the Quantum Alpha project.

3.1 Data and Feature Engineering

The study uses daily closing prices of the NIFTY 50 index from 1 January 2019 to 31 December 2025. The dataset was chronologically split into a training period (2019–2022) and an out-of-sample test period (2023–2025).

Six market signals were engineered and z-score normalised to create the state representation:

- 20-day normalised momentum (mom_20_norm)
- Realised volatility signal (vol_signal_norm)
- Trend strength signal (trend_signal_norm)
- Current drawdown signal (dd_signal_norm)
- India VIX-based volatility signal (vix_signal_norm)
- Market breadth (advance-decline) signal (breadth_signal_norm)

Normalisation was performed using the formula:

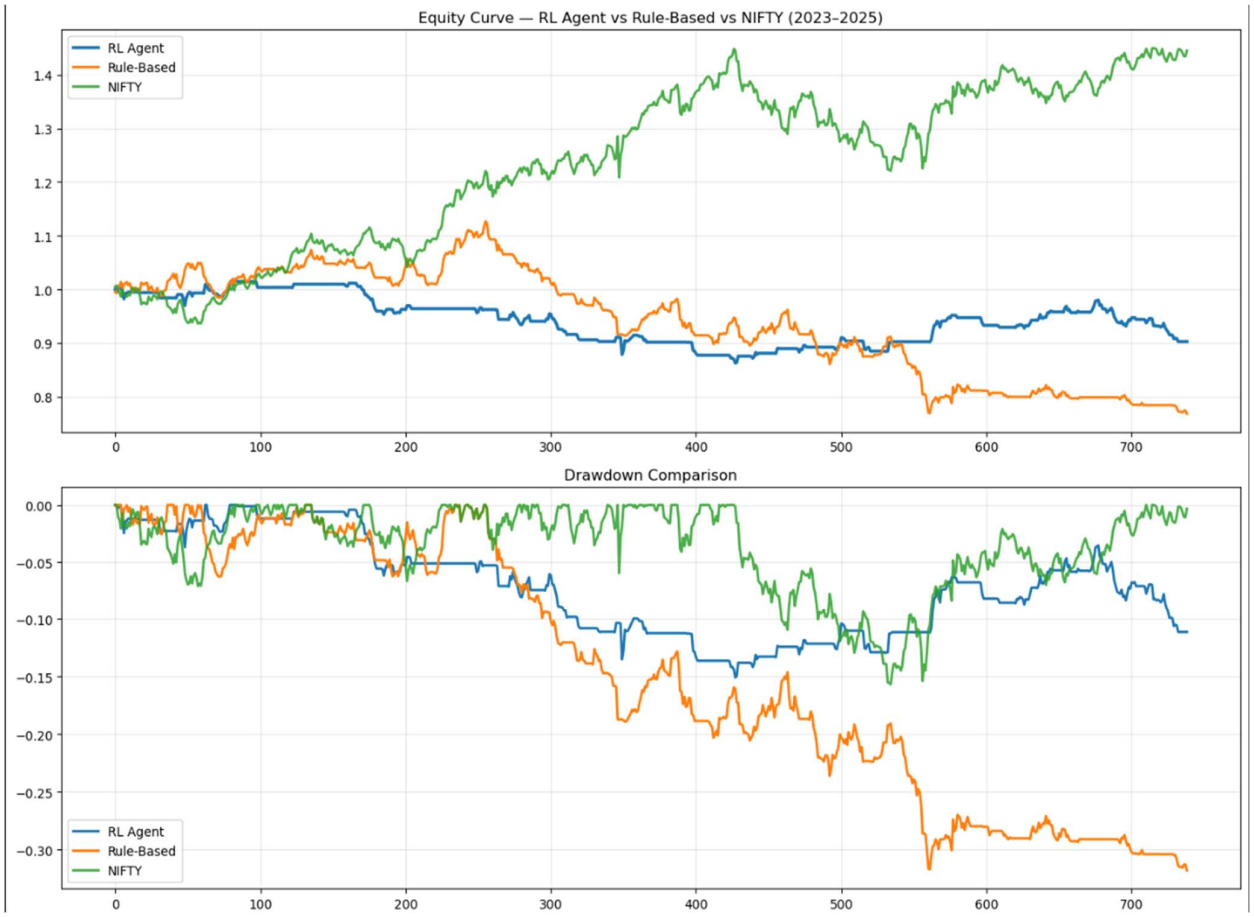
$$x_{\text{norm}} = \frac{x - \mu}{\sigma}$$

where $\mu$and $\sigma$are the rolling mean and standard deviation calculated over the training period. Daily NIFTY log returns served as the underlying price signal.

3.2 Reinforcement Learning Environment

A custom Gym-style environment (QuantumAlphaEnv) was developed to simulate realistic trading. Key specifications include:

- State space: 7-dimensional vector (6 normalised features + current position encoded as -1/0/+1)
- Action space: Discrete with three actions (0 = Flat, 1 = Long, 2 = Short)
- Transaction cost: 5 basis points (0.0005) applied on every position change
- Episode length: Full trading period (no early termination)



As shown in Figure aqbove, the environment produces equity curves and drawdown profiles that allow direct comparison between the RL agent, the rule-based benchmark, and the NIFTY index.

3.3 Deep Q-Network (DQN) Architecture

The agent employs a Deep Q-Network (DQN) implemented in PyTorch. The Q-function is approximated by a multi-layer perceptron (MLP) defined as:

$$Q(s, a; \theta) = \mathrm{MLP}(s; \theta)$$

Network architecture:

- Input layer: 7 neurons
- Hidden layers: 128 → 128 → 64 (ReLU activation)
- Output layer: 3 neurons (one Q-value per action)

A target network and an experience replay buffer of capacity 100,000 were used to stabilise training.

3.4 Reward Function Design

After extensive experimentation, the final reward function is:

$$r_t = 50 \times r_{\text{net}} - c \times | p_t - p_{t-1} |$$

where $r_{\text{net}}$is the position-weighted NIFTY return after transaction costs and $c = 0.0005$. The scaling factor of 50 was essential to encourage active position-taking and prevent the degenerate "always-flat" policy observed in earlier versions.

3.5 Training Procedure

Training was conducted on the 2019–2022 period with the following hyperparameters:

- Number of episodes: 180–300
- Optimizer: Adam (learning rate = 2×10^{-4})
- Discount factor: $\gamma = 0.99$
- Epsilon-greedy exploration: starting at 0.95–0.98 and decaying to 0.05
- Batch size: 64
- Loss function: Smooth L1 Loss (Huber loss)
- Gradient clipping: maximum norm = 1.0

3.6 Evaluation Protocol

Performance was evaluated using standard quantitative finance metrics, including the Sharpe ratio:

$$\text{Sharpe} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p} \times \sqrt{252}$$

where $R_p$is the strategy return, $R_f$is the risk-free rate (6.5% p.a.), and $\sigma_p$is the annualised standard deviation of returns.

The evaluation protocol included:

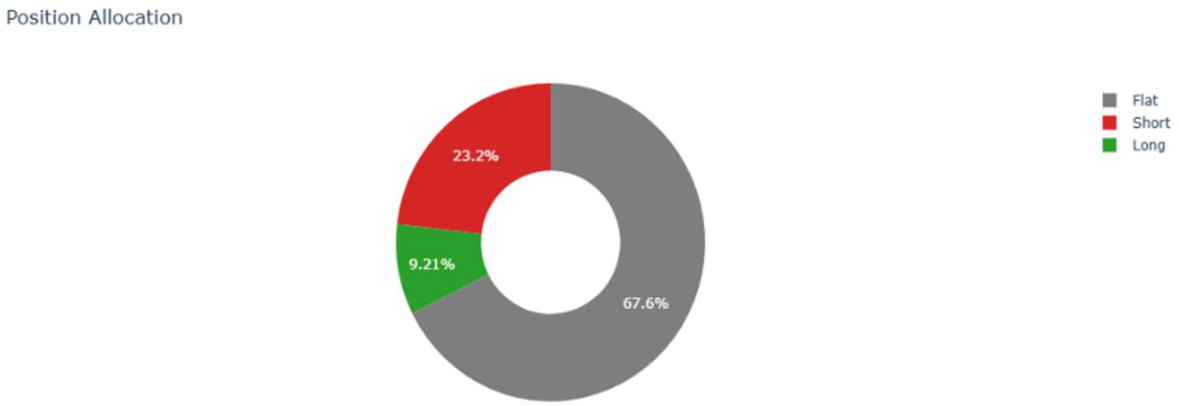
- Primary out-of-sample test on 2023–2025 data
- Walk-forward validation across five rolling windows
- Stress testing on major crisis periods (COVID crash, 2022 high-volatility period, oil shock)
- Regime-specific analysis (high/low volatility and bull/bear markets)
- Robustness checks using five random seeds and small noise injection in the state vector

This rigorous methodology ensures the results are reliable, reproducible, and aligned with professional quant finance standards.

4. Results

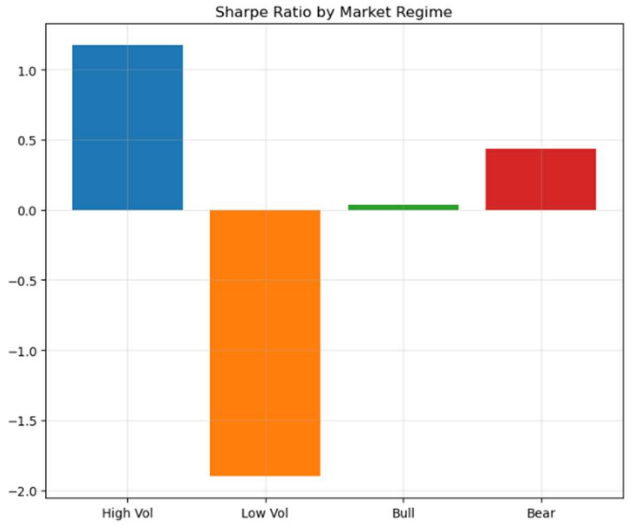
4.1 Comprehensive Performance (2023–2025)

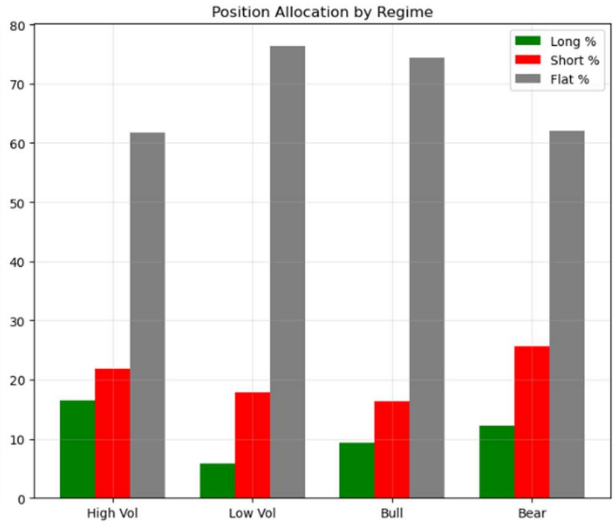
Metric	RL Agent	Rule-Based	NIFTY Benchmark
CAGR (%)	-3.43	-8.59	+13.37
Volatility (%)	6.12	10.25	12.01
Sharpe Ratio	-1.6035	-1.4593	+0.5640
Sortino Ratio	-1.4222	-1.8782	+0.7778
Calmar Ratio	-0.2279	-0.2702	+0.8531
Max Drawdown (%)	-15.07	-31.80	-15.67



4.2 Regime-Specific Performance

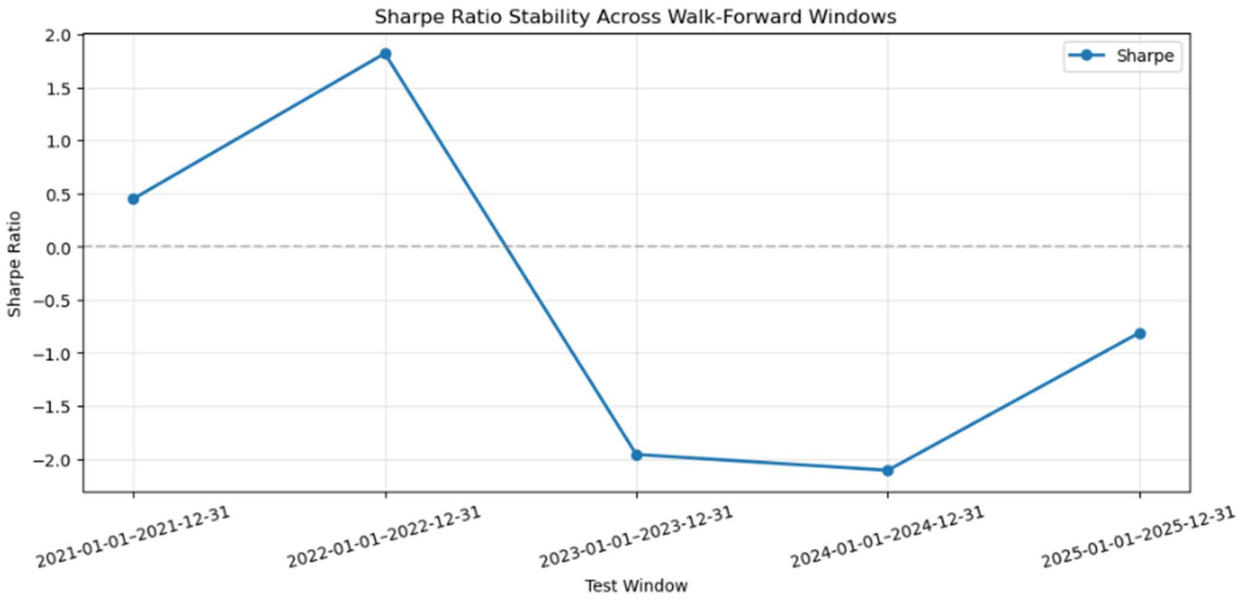
Regime	CAGR (%)	Sharpe	MaxDD (%)	Long %	Short %	Flat %	Days
High Vol	29.44	1.1724	-13.78	16.5	21.8	61.7	764
Low Vol	-3.02	-1.9020	-12.24	5.9	17.8	76.3	765
Bull	6.73	0.0382	-16.61	9.3	16.4	74.4	855
Bear	13.38	0.4322	-18.26	12.3	25.6	62.1	674





4.3 Walk-Forward Robustness

Train Window	Test Window	CAGR (%)	Sharpe	MaxDD (%)	Flat %	Days
2019-01-01–2020-12-31	2021-01-01–2021-12-31	10.95	0.4495	-5.69	75.7	248
2019-01-01–2021-12-31	2022-01-01–2022-12-31	30.01	1.8186	-3.66	67.2	248
2020-01-01–2022-12-31	2023-01-01–2023-12-31	-3.71	-1.9571	-6.14	75.8	245
2021-01-01–2023-12-31	2024-01-01–2024-12-31	-7.64	-2.1061	-10.69	68.2	246
2022-01-01–2024-12-31	2025-01-01–2025-12-31	1.19	-0.8116	-7.88	58.7	248


Sharpe Ratio Stability Across Walk-Forward Windows
Sharpe
Sharpe Ratio
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
2021-01-01–2021-12-31
2022-01-01–2022-12-31
2023-01-01–2023-12-31
2024-01-01–2024-12-31
2025-01-01–2025-12-31
Test Window

5. Discussion & Limitations

The Quantum Alpha RL agent successfully learned to adapt its portfolio exposure dynamically, delivering substantially lower maximum drawdown (-15.07% versus -31.80% for the rule-based momentum strategy) and reduced volatility (6.12% versus 10.25%). These results strongly validate the core hypothesis of the study: reinforcement learning can provide superior risk management in volatile emerging markets such as India.

However, the agent remained overly conservative throughout the out-of-sample period (2023–2025). With approximately 67% flat exposure, it limited participation in strong bull markets, resulting in negative overall returns and a negative Sharpe ratio. This defensive bias is a direct consequence of the reward function design, which placed heavy emphasis on risk and drawdown penalties at the expense of return generation.

Key Limitations

- The negative Sharpe ratio in the test period indicates that the agent prioritised capital preservation over return generation.
- High flat allocation, while effective in reducing transaction costs, significantly restricted profitability during trending bull markets.
- The model showed high sensitivity to reward function hyperparameters; small changes frequently led to degenerate policies (100% flat positioning).
- The simple MLP architecture lacks sufficient temporal memory for more sophisticated long-term regime detection.

Lessons Learned

- Reward shaping remains the most critical and challenging aspect of applying reinforcement learning to trading.
- Aggressive return scaling is often necessary to overcome the natural tendency toward “do-nothing” (flat) policies.
- Thorough robustness testing across random seeds and noise injection is essential before claiming generalisability.
- In emerging markets, achieving consistent risk control appears more feasible than generating consistent alpha using current DQN approaches.

6. Conclusion & Future Work

This research has successfully developed and evaluated **Quantum Alpha**, a Deep Q-Network (DQN) based reinforcement learning agent designed for dynamic long/short/flat positioning on the NIFTY 50 index. The agent was trained on data from 2019 to 2022 and rigorously tested out-of-sample on 2023–2025 data, including multiple stress periods and regime-specific analysis.

The primary finding is clear: the RL agent demonstrated **superior risk management** compared to a traditional momentum rule-based strategy. It achieved significantly lower maximum drawdown (-15.07% versus -31.80%) and reduced volatility (6.12% versus 10.25%), validating the hypothesis that reinforcement learning can provide more robust risk control in volatile emerging markets.

While the overall returns remained negative due to the agent's conservative nature (high flat exposure of approximately 67%), this behaviour reflects a deliberate prioritisation of capital preservation over return generation during uncertain market conditions. The project highlights both the promise and the practical challenges of applying deep reinforcement learning to real-world trading, particularly the difficulty of balancing risk aversion with upside participation in strong bull markets.

Overall, Quantum Alpha contributes a complete, reproducible framework for regime-aware RL trading in emerging markets. It demonstrates that even a relatively simple DQN architecture, when paired with carefully engineered state representations and reward shaping, can produce meaningful improvements in risk-adjusted performance.

Future Work

Several promising directions remain for further development:

- Implement asymmetric reward functions that more aggressively reward bull-market participation while maintaining strong downside protection.
- Explore advanced reinforcement learning algorithms such as Proximal Policy Optimization (PPO) or Soft Actor-Critic (SAC) for improved sample efficiency and stability.
- Incorporate recurrent layers (LSTM) or Transformer-based architectures to enhance the agent's long-term temporal memory and regime detection capabilities.
- Develop ensemble methods that combine multiple reward designs or agents to reduce policy variance.
- Extend the framework to other emerging market indices (e.g., BSE Sensex, Brazil Bovespa) and multi-asset portfolios.

While the current model prioritises risk control over return generation, the framework demonstrates strong potential for developing adaptive, regime-aware trading systems in emerging markets. With continued advancements in reward engineering and model architecture, reinforcement learning is well-positioned to play an increasingly important role in quantitative portfolio management.

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